Cloud Computing Course

Course n°1: Introduction to Cloud Computing

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👋 Welcome to the Cloud Computing Course

📅 Structure

- **6 sessions × 4 hours** (2h theory + 2h hands-on)
- **Resources & deliverables** shared after each session

📝 Evaluation

- Final exam format **TBD**
- Likely: **Project**

💬 Interaction

- Ask questions anytime!
- I may not always have the answer → I'll follow up
- I can be wrong → feel free to correct me 🙌

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Aim of this session

- Discover the history of cloud computing.
- Learn why the cloud is increasingly used by businesses today.
- Understand the overall characteristics of the cloud.
- Manipulate and test the deployment of cloud resources.

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Contenu

Introduction to Cloud Computing

1. Before the Cloud
2. The Emergence of Cloud Computing
3. The Current State of CLoud Computing
4. The 5 Essentials characteristics
5. The 3 Service Models
6. The 4 Deployment Models
7. Hands-on: Azure

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Before the Cloud

On-Premise

Before the Cloud: On-Premise (1/2)

Definition

- All IT resources hosted on the organization's premises
- Company owns and manages servers, storage, networking
- Requires in-house IT staff and physical infrastructure

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🏙️ Before the Cloud: On-Premise (2/2)

✅ Advantages

- Full control over infrastructure
- High level of security (data stays on-site)
- Customization possible

❌ Disadvantages

- High upfront investment
- Long provisioning time for new resources
- Limited scalability (hardware-bound)
- Inefficient: hardware bought for peak demand, idle most of the time ~10% capacity

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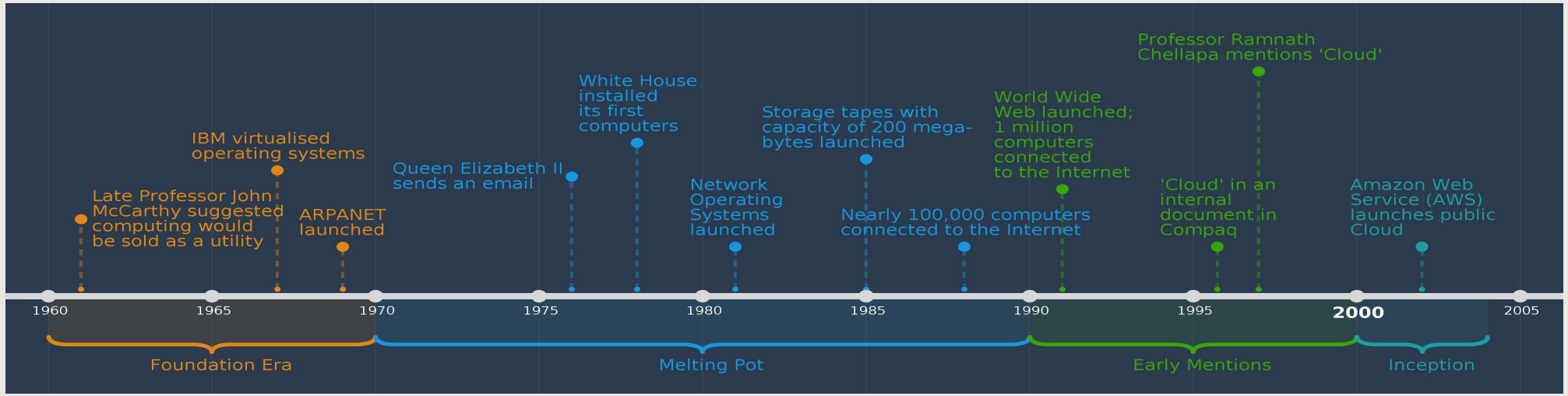
The Emergence of

Cloud Computing

The Emergence of Cloud Computing

Context

- **Widespread Internet Access**: The proliferation of broadband and global connectivity.
- **Virtualization Technologies**: Enabled efficient resource utilization and isolation.
- **Global Data Centers**: Providers began offering scalable infrastructure services.



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What is a Data Center? (1/2)

Definition

- A **data center** is a facility used to house **computer systems and associated components** such as servers, storage systems, and networking equipment.
- Provides **reliable power, cooling, connectivity, and security** to support IT infrastructure.

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What is a Data Center? (2/2)

Four-tier of Data Center

Tier	Key Features	Uptime
Tier I	Basic infrastructure, minimal redundancy with UPS and 24/7 cooling	99.671% (~28-29 h/yr)
Tier II	Adds extra backup power & cooling to reduce downtime	99.741% (~22.7 h/yr)
Tier III	Fully redundant, maintenance without shutdown	99.982% (~1.6 h/yr)
Tier IV	Fault-tolerant, multiple independent backups, failures have no impact	99.995% (~30 min/yr)

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What is Virtualization? (1/2)

Definition

- Virtualization is the creation of **virtual versions of physical computing resources** (servers, storage, networks) using software.
- Allows **multiple virtual machines (VMs)** to run on a single physical host.

Key Concept

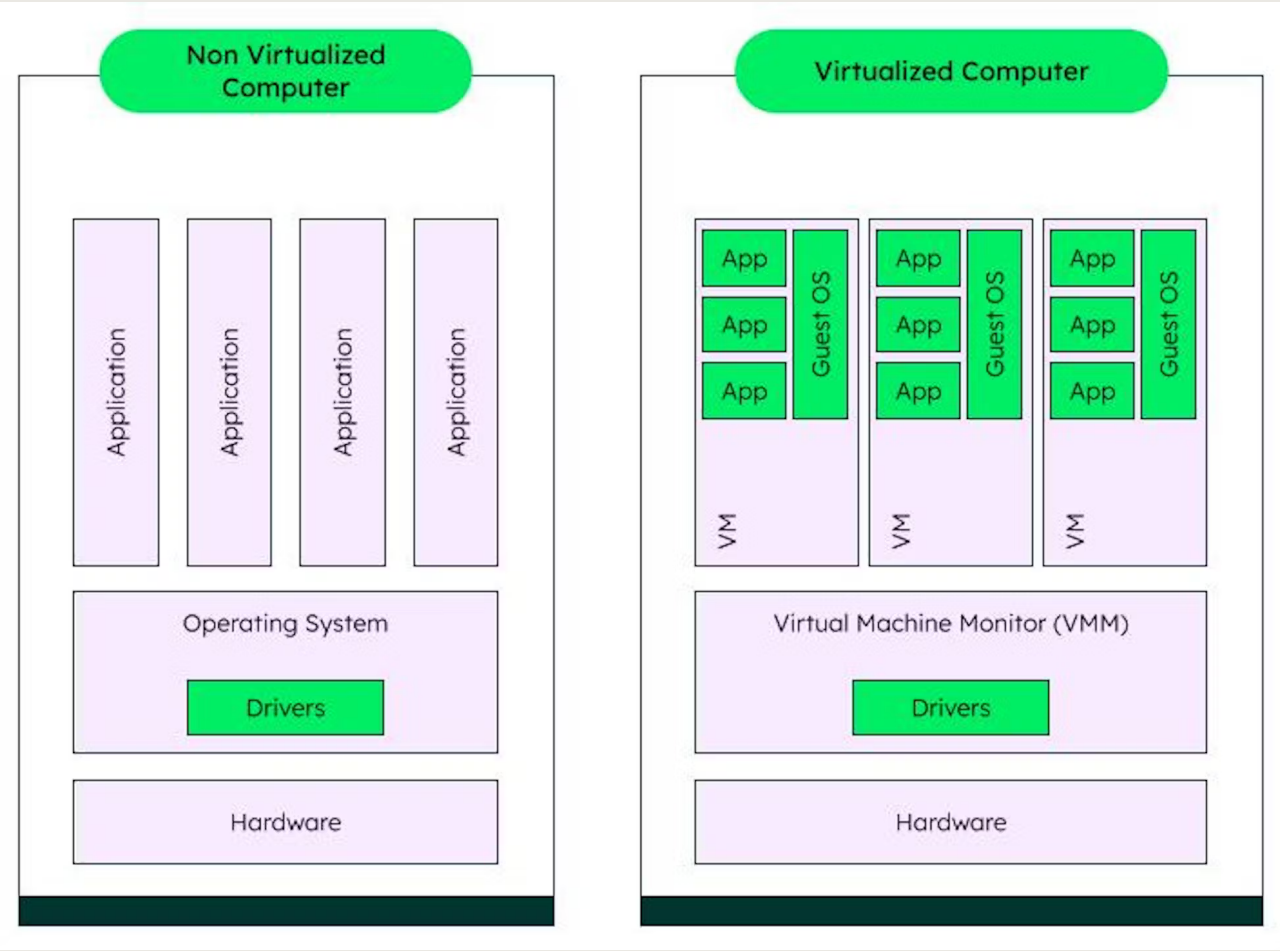
- One physical machine → multiple isolated environments
- Enables better utilization, flexibility, and easier management of IT resources.

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What is Virtualization? (2/2)

How it works

1. A **hypervisor** sits between the hardware and virtual machines.
2. Each VM runs its **own operating system** and applications independently.
3. Virtual resources (CPU, memory, storage, network) are **shared and managed efficiently** by the hypervisor.



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The Current State of

Cloud Computing

What is Cloud Computing?

Cloud computing is a model that enables
ubiquitous, convenient, on-demand access
to a *shared pool of configurable resources*
(networks, servers, storage, applications, services)
that can be *rapidly provisioned and released*
with *minimal management effort* or *provider interaction*.

— *National Institute of Standards and Technology (NIST)*

The 5 Essentials characteristics

of Cloud Computing

The 5 Essential characteristics (1/3)

- **On-demand self-service**
  - Users can provision computing resources (e.g., server time, storage) automatically.
  - No human interaction needed with the provider.
- **Broad network access**
  - Resources available over the network.
  - Accessible via standard mechanisms (e.g., mobile phones, tablets, laptops, workstations).

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The 5 Essential characteristics (2/3)

- **Rapid elasticity**
  - Resources scale up or down quickly, automatically or not.
  - To users, resources appear unlimited and available anytime in any quantity.
- **Measured service**
  - Automatic monitoring, control, and optimization of resources.
  - Usage is metered (e.g., storage, processing, bandwidth, user accounts).
  - Transparency for both provider and consumer.

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The 5 Essential characteristics (3/3)

- **Resource pooling**
  - Shared resources in a multi-tenant model.
  - Physical and virtual resources dynamically allocated based on demand.
  - Location independence: users don't know or control exact resource locations (may specify region/datacenter).
  - Available for many types of resources: storage, processing, memory, bandwidth, ...

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The 3 Service Models

of Cloud Computing

The 3 Service Models (1/3)

Software as a Service (SaaS)

- Ready-to-use applications, accessed via the Internet
- No management of infrastructure (servers, storage, OS)
- Limited to user-level configuration

Examples:

- Gmail, Microsoft 365, Dropbox



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The 3 Service Models (2/3)

Platform as a Service (PaaS)

- Environment to build and deploy applications
- Provider offers languages, frameworks, libraries, services
- No management of servers, network, or storage
- Control limited to the application and its runtime environment

Examples:

- Google App Engine, Heroku, AWS Elastic Beanstalk

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The 3 Service Models (3/3)

Infrastructure as a Service (IaaS)

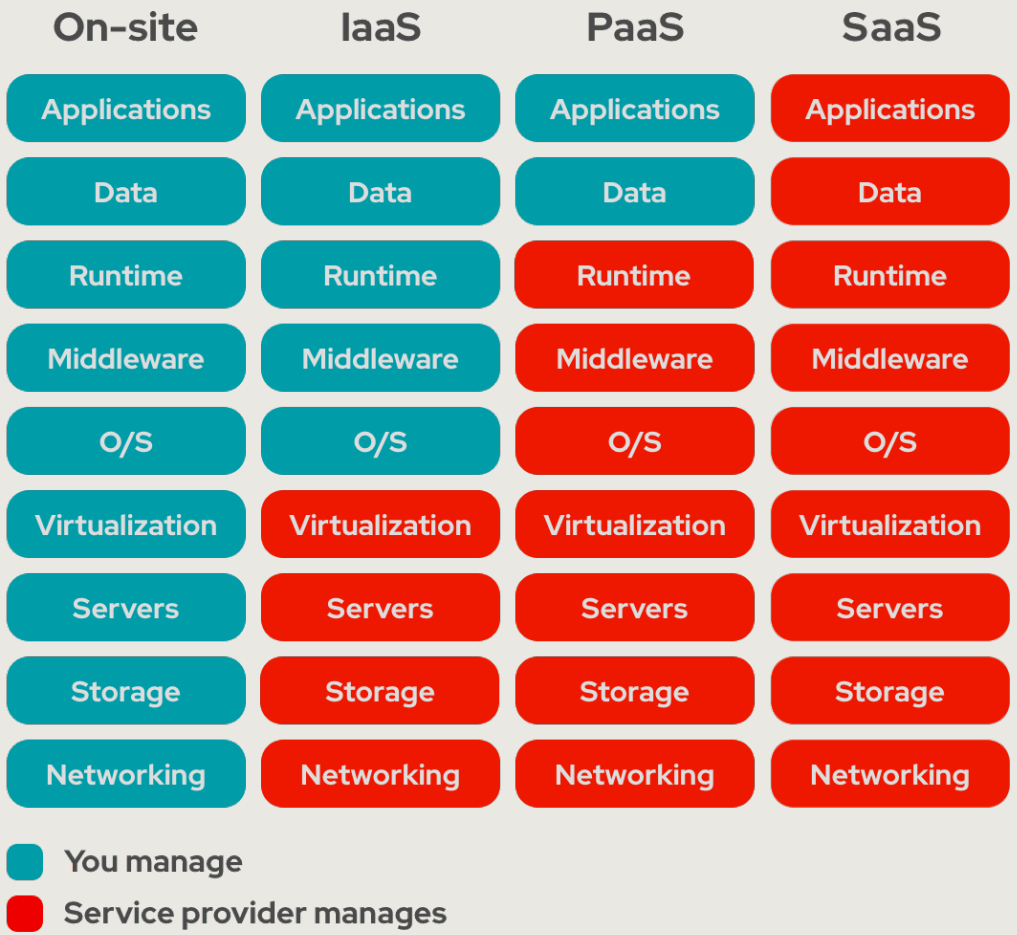
- Access to fundamental computing resources: VMs, storage, network
- Full control over OS and deployed applications
- No physical server management

Examples:

- Amazon Web Services, Microsoft Azure, Google Compute Engine

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In short



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The 4 Deployment Models

of Cloud Computing

The 4 Deployment Models (1/4)

Public Cloud

Definition from NIST

The cloud infrastructure is provisioned for open use by the general public. It may be owned, managed, and operated by a business, academic, or government organization, or some combination of them. It exists on the premises of the cloud provider.

In short

- Available to the general public
- Owned & operated by a provider (business, academic, or government)
- Infrastructure hosted **on provider premises**

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The 4 Deployment Models (2/4)

Private Cloud

Definition from NIST

The cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers (e.g., business units). It may be owned, managed, and operated by the organization, a third party, or some combination of them, and it may exist on or off premises.

In short

- Dedicated to a **single organization**
- Can be managed by the organization itself or a third party
- Hosted **on-premises** or in a provider's datacenter

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The 4 Deployment Models (3/4)

Community Cloud

> **Definition from NIST**
> The cloud infrastructure is provisioned for exclusive use by a specific community of consumers from organizations that have shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be owned, managed, and operated by one or more of the organizations in the community, a third party, or some combination of them, and it may exist on or off premises.

In short

- Shared by a **specific group of organizations**
- Designed around **common requirements**
  - Security
  - Compliance
  - Mission goals
- Can be on-premises or hosted externally

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The 4 Deployment Models (4/4)

Hybrid Cloud

Definition from NIST

The cloud infrastructure is a composition of two or more distinct cloud infrastructures (private, community, or public) that remain unique entities, but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load balancing between clouds)

In short

- Combination of **two or more cloud models** (public, private, community)
- Interconnected for **data and application portability**
- Typical use case: **cloud bursting** (overflow traffic handled by public cloud)

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Summary

Model	Users	Ownership / Management	Location
Public	General public	Cloud provider	Provider's datacenter
Private	One single organization	Organization or 3rd party	On-premises / hosted
Community	Group of organizations	One or more in the community / 3rd party	On-premises / hosted
Hybrid	Mix of models (public/private/community)	Shared responsibility	Combination of locations

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Cloud Computing: Pros & Cons

✅ Advantages

- **Cost efficiency**: Pay-as-you-go, reduced upfront investment
- **Scalability & Elasticity**: Adapt resources instantly
- **Accessibility**: Anywhere, anytime, on any device
- **Maintenance offloaded**: Provider manages infrastructure
- **Reliability & Redundancy**: High availability, disaster recovery
- **Faster deployment**: Quick provisioning of services

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Cloud Computing: Pros & Cons

❌ Disadvantages

- **Security & Privacy** concerns: Data outside the organization
- **Dependence on Internet**: No access without connectivity
- **Unpredictable costs**: Hard to estimate with scaling workloads
- **Vendor lock-in**: Difficult migration between providers
- **Compliance challenges**: Regulations vary across regions
- **Environmental impact**: Large-scale data centers consume significant energy and contribute to carbon emissions

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The Main Actors

of Cloud Computing

Main Actors







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Learning about each Cloud Providers

You can find a lot a free online learning for each Cloud Providers

Azure

- Learning Subscription: https://azure.microsoft.com/en-us/free/students
- Course: https://learn.microsoft.com/en-us/training/courses/az-900t00
- Certification (Optional): https://learn.microsoft.com/en-us/credentials/certifications/azure-fundamentals/?practice-assessment-type=certification

GCP

- Learning Subscription: https://www.cloudskillsboost.google/subscriptions
- Course: https://www.cloudskillsboost.google/paths?locale=en&utm_campaign=evergreen&utm_medium=website&utm_source=cgc
- Certification (Optional): https://cloud.google.com/learn/certification?hl=en

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Thank you !

Let's take a break before the practice session!